

Logic Quick Reference Sheet

1. Basic Operators

NOT ($\neg p$) $\rightarrow$ flips the value

$\neg 1 = 0$, $\neg 0 = 1$

AND ($p \wedge q$) $\rightarrow$ true only if both are true:

p	q	$p \wedge q$
0	0	0
0	1	0
1	0	0
1	1	1

OR ($p \vee q$) $\rightarrow$ true if at least one is true:

p	q	$p \vee q$
0	0	0
0	1	1
1	0	1
1	1	1

2. Implication ($p \rightarrow q$)

"If p then q" $\rightarrow$ false only when p = 1 and q = 0.

p	q	$p \rightarrow q$
0	0	1
0	1	1
1	0	0
1	1	1

Tip: $p \rightarrow q \equiv (\neg p \vee q)$

3. Contrapositive

$p \rightarrow q \equiv \neg q \rightarrow \neg p$

Example: 'If it rains $\rightarrow$ ground is wet' is the same as 'If ground is not wet $\rightarrow$ it didn't rain'.

4. Common Tips

- AND with 0 $\rightarrow$ always 0.
- OR with 1 $\rightarrow$ always 1.
- Implication to 1 ($x \rightarrow 1$) $\rightarrow$ always true.

- Implication from 0 ($0 \rightarrow y \rightarrow$ always true).